ModelIt!

Student Activity Pack

How Does Water Shape the Land?

The Power of Weathering and Erosion

Grade 3 | NGSS 3-ESS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever seen a stream or river carving a path through dirt or mud? What did it look like?

2 How do you think the Grand Canyon — one of the biggest holes on Earth — was made?

3 Draw what you think happens to soil on a hillside when it rains really hard.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Water Flow | Erosion Strength | Land Shape Change

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gentle Stream

Water Flow set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Flooding River

Water Flow set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Steady Rain

Water Flow set to medium

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Slope Steepness — How steep the hill or mountain is — steeper slopes cause water to flow faster and erode more

Vegetation Cover — How many plants and trees cover the ground — plant roots hold soil in place and slow erosion

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can something as soft as water carve giant canyons and move huge boulders? Use evidence from your simulation to support your answers.

1 How can something as soft as water be powerful enough to carve through solid rock?

2 How did your model help you understand why some places on Earth look the way they do?

8 STEM Challenge: Design an Erosion-Proof Hillside

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YOUR ENGINEERING MISSION

Design a hillside model that resists erosion by using plants, rocks, or barriers to slow water flow and prevent soil from washing away.

THE SCENARIO

A town built on a hillside is losing soil every time it rains. The town council hired your engineering team to design a solution that prevents erosion and keeps the hillside stable. Use what you learned about water flow and erosion to protect the town!

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO BUILD YOUR HILLSIDE — SOIL, SAND, GRAVEL, CLAY?

2. WHAT WILL YOU PLANT OR PLACE ON THE HILL TO SLOW EROSION — GRASS, ROCKS, TERRACES?

3. HOW WILL YOU TEST YOUR DESIGN — POURING WATER FROM A CUP, USING A SPRAY BOTTLE?

4. HOW WILL YOU MEASURE HOW MUCH SOIL WASHED AWAY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Geologists study rocks, landforms, and Earth's processes including weathering and erosion. They earn $55,000-$105,000/year.